Week 1. Imp. GAs.

11) Your portfolio consists of three blue bonds: each blue bond has a face value 100 and a probability of default p_b = 10%. The correlation between blue bonds is $\gamma = 0$. Through securitization, you are able to create three new bonds with face values: a green bond, a yellow bond and a red bond. When none of the bonds default all three new bonds are paid face value of 100. If any one of the blue bond defaults, the green bond and the yellow bond are paid the face value of 100 while the red bond suffers the loss and is paid 0. If two blue bonds default, only green bond is paid the face value, other two are paid 0. In case of all three blue bonds defaulting, none of the new bonds pays off. We are interested in calculating the fair value of the green bond, yellow bond and red bond. (Note: Write your answers up to three decimal points.)

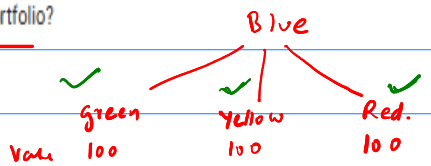
What is the probability that all three blue bonds default? = $0.1 \times 0.1 \times 0.1 = 0.001$ i.e. 0.1%

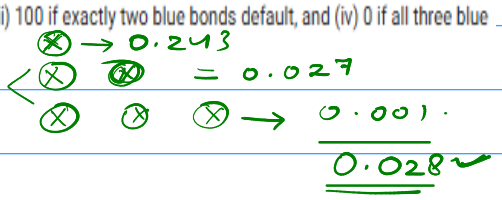
12) What is the probability that none of the blue bonds default? = $0.9 \times 0.9 \times 0.9 = 0.81 \times 9 = 0.729$

13) What is the probability that exactly one of blue bonds defaults? = $3 \times 0.1 \times 0.9 \times 0.9 = 3 \times 0.081 = 0.243$

14) What is the probability that exactly two blue bonds default? = $3 \times 0.1 \times 0.1 \times 0.9 = 3 \times 0.009 = 0.027$

15) Your portfolio has the following values: (i) 300 if none of the blue bonds default, (ii) 200 if exactly one blue bond defaults, (iii) 100 if exactly two blue bonds default, and (iv) 0 if all three blue bonds default. What is the expected value of your portfolio?





P (green defaults, when all 3 default) = 0.001 ∴ P (green not default) = 1 − 0.001 = 0.999

∴ Expected payoff (green) = 0.999×100 = 99.9

P (yellow defaults, when all 3 default or when 2 default) = 0.001 + 0.027 = 0.028.

∴ P (yellow not defaulting) = 1 − 0.028 = 0.972

∴ Expected Payoff (yellow) = 0.972×100 = 97.2.

P (Red default) = P(all default) + P(2 default) + P(one default) = 0.001 + 0.027 + 0.243 = 0.271

∴ P (Red not defaulting) = 1 − 0.271 = 0.729

Expected payoff (Red) = 0.729×100 = 72.9

∴ Total Payoff, expected = 72.9 + 97.2 + 99.9 = 270.0

|| ✱ Imp. to remember this. Ques might come.

16) What is the variance of your portfolio? = $E(X^2) - [E(X)]^2$ 270.

P(300) = 0.729 $(0.9 \times 0.9 \times 0.9)$

P(200) = $3 \times 0.1 \times 0.9 \times 0.9$ (only one defaulting) = 0.243.

P(100) = $3 \times 0.1 \times 0.1 \times 0.9$ = 0.027

P(0) = $3 \times 0.1 \times 0.1 \times 0.1$ = 0.001

∴ $E(X^2)$ = $0.729 \times 90000 + 0.243 \times 40000 + 0.027 \times 10000 + 0$.

= 65610 + 9720 + 270 = 75600

∴ Variance = 75600 − 270^2 = 2700

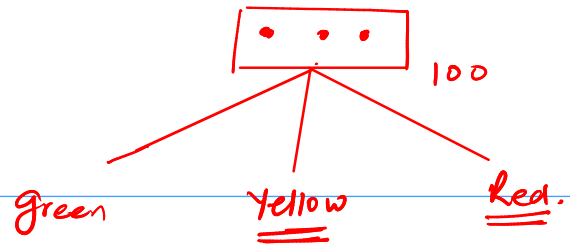
∴ $\sigma = \sqrt{2700}$

18) What is the expected payoff of a green bond across all states? =

green will default only if all 3 will default P = 0.001

∴ P (green will not default) = 1 − 0.001 = 0.999.

∴ E(Payoff green) = 100×0.999 = 99.9



0.1

If any bond 'defaults' → Red will suffer. (default)

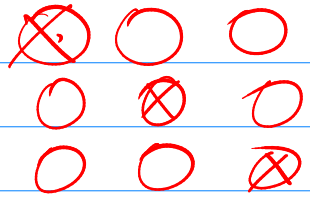
If any two bonds 'default' → Yellow + Red will default

Green will default → All 3 will default.

P (green default) = $0.1 \times 0.1 \times 0.1 = 0.001$

'Green' will be ranked higher.

One default.



$= 3 \times 0.1 \times 0.9 \times 0.9$.

$$1 + r_{0.5} = (1+r)^{12}$$

$$1.05 = (1+r)^{12}$$

$$\ln 1.05 = 12 \ln(1+r)$$

$$\therefore \ln(1+r) = \frac{\ln 1.05}{12} = 0.707$$ (exact)

$$1 + r = e^{0.707}$$

$$\therefore r = e^{0.707} - 1$$

for. 5000 every year for 20 years
rate of. 20%.

$$A = \frac{5000}{1.2} + \frac{5000}{1.2^2} + \cdots + \frac{5000}{1.2^{20}}$$

$$= \frac{C}{r}\left[1 - \frac{1}{(1+r)^t}\right]$$

$$\frac{A}{1.2} = \frac{5000}{1.2^2} + \cdots + \frac{5000}{1.2^{21}}$$

$$A - \frac{A}{1.2} = \frac{5000}{1.2} - \frac{5000}{1.2^{21}}$$

19) What is the expected payoff of a yellow bond across all states?

Yellow will default only if either all 3 default or only 2 default.

$\therefore P(\text{yellow default}) = 0.001 + 3 \times 0.1 \times 0.1 \times 0.9 = 0.001 + 0.027 = 0.028$

$\therefore P(\text{Yellow not defaulting}) = 1 - 0.028 = 0.972$

$\therefore$ Expected Payoff (Yellow) $= 0.972 \times 100 = 97.2$.

20) What is the expected payoff of a red bond across all states?

$P(\text{Red default}) = P(\text{all default or only 2 default or only one default})$.

$= 0.001 + 0.027 + 3 \times 0.1 \times 0.9 \times 0.9 = 0.028 + 0.243 = 0.271$

$\therefore P(\text{Red not defaulting}) = 1 - 0.271 = 0.729$.

$\therefore$ Expected Payoff (Red) $= 0.729 \times 100 = 72.9$.

Week 2 : Imp Concepts.

1. $1 + r_t = (1 + r)^t$, $r_t \rightarrow$ Compound interest , $r \rightarrow$ Simple interest (annualised returns), $t \rightarrow$ time periods.

- **Qn**: How much would an investment project which costs 100 today, and earns 10% each year be worth in 12 years?

$A = P(1+r)$

$A = 100(1+0.1)^{12}$.

$1 + r_t = (1+0.1)^{12} = (1.1)^{12} = 3.138 \quad \therefore r_t = 2.138$.

$\therefore$ ✓ $X = 100 + 100 \times 2.138 = 100 + 213.8 = 313.8$ ✓

- **Qn**: What constant two 1-year interest rate (r) would give a two-year interest rate of 50%?

$\therefore 1 + 0.5 = (1+r)^2 \quad \therefore (1+r)^2 = 1.5 \quad \therefore 1 + r = \sqrt{1.5} \quad \therefore r = 1.2247 - 1 = 0.2247$ or 22.5%.

- **Qn**: At an interest rate of 3% per years, how many years would it take to double one's money?

$\therefore 1 + 1 = (1 + 0.03)^t \quad \therefore 1.03^t = 2 \quad \therefore t = \frac{\log 2}{\log 1.03} = \frac{0.30}{0.0128} = 23.44$

- **Qn**: If the overall interest rate is 5% per year, what is the rate of return over half a year?

$(1+r)^2 = 1 + 0.05 \quad \therefore r = \sqrt{1.05} - 1 = 0.0246 = 2.46\%$

- Application in corporate context: Given that Project Y will return 1 million in 5 years, how much should one be willing to pay to undertake this project today? 'interest rate = 20%'

$\therefore PV = \frac{1000000}{(1.2)^5} = 783526.66$

- **Qn**: Suppose that a project costs 90.91 and will give back 100 next year. If the standard rate of return in the economy were 12%, would this project be a good deal?

$PV = \frac{100}{1.12} = 89.28 < 90.91$

$\therefore$ not a good deal.

- Consider a project that generates 10 in one year and 8 in five years. Find PV? $r = 20\%$.

Ans $PV = \frac{10}{1.2} + \frac{8}{(1.2)^5} = 8.33 + 3.21 = 11.54$

- Suppose you have to pay 12 units today for a project which pays out 10 units 1 year later and 8 units in 5 years. The rate of interest is 5% per annum. Find NPV.

$NPV = PV - \text{present cost} = \frac{10}{1.05} + \frac{8}{(1.05)^5} - 12 = 9.52 + 6.27 - 12 = 15.79 - 12 = 3.79.$

- **Qn**: Consider a project which costs 900 units today, yields 200 units per year for two years, then 400 units per year for next two years and finally requires a cleanup expense of 100 the next year.
- The interest rate is 5 per cent per year. Should one take up the project?

payments: $-900 \quad 200 \quad 200 \quad 400 \quad 400 \quad -100$

$\therefore NPV = -900 + \frac{200}{1.05} + \frac{200}{1.05^2} + \frac{400}{1.05^3} + \frac{400}{1.05^4} - \frac{100}{1.05^5}$

$= -900 + 190.5 + 181.4 + 345.5 + 329 - 78.5$

$= 1046.4 - 978.5 = +ve.$

Yes.

Imp ★ fourth. real interest rate, interest rate, inflation.

$(1+\rho) = \frac{1+r}{1+i}$ $\rho \to$ real interest rate, $r \to$ interest rate, $i \to$ inflation.

- **Qn**: At a constant interest rate of 10%, how much money does one need to invest today to receive the same amount, 20 units per year, starting next year, forever?] perpetuity →

$x = \frac{20}{1.1} + \frac{20}{1.1^2} + \frac{20}{1.1^3} + \dots$

$\therefore \frac{x}{1.1} = \frac{20}{1.1^2} + \frac{20}{1.1^3} + \dots$

$\therefore x - \frac{x}{1.1} = \frac{20}{1.1} \quad \therefore x = \frac{20}{1.1}\left(\frac{1.1}{0.1}\right) = \frac{20}{0.1} = 200$

formula $PV = \frac{C}{1+r}\left[\frac{1+r}{1+r-1}\right] = \frac{C}{r}$

★ $PV_0 = \frac{C}{r}$ — imp. (for Perpetuity)

- What happens if instead of the same amount of cash every period, the cash flow increases over time?
- The growing perpetuity formula allows for a constant rate per period, provided it is less than the interest rate.
- **Qn**: What is the value today of a cash flow stream that pays 20 units next year, grows at a rate of 5 percent, and faces a cost of capital of 10%? meaning from first year from next year.

grow $g\%$.

$PV = \frac{C}{r-g}$

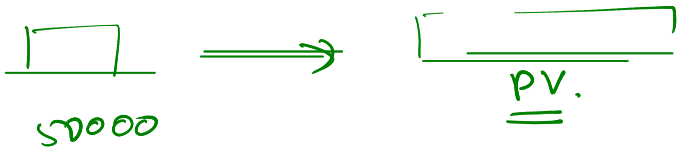
$\frac{20}{0.1-0.05} = \frac{2000}{0.05}$

$\frac{C}{r}, \quad \frac{C}{r-g}.$

For annuity, $PV_0 = \frac{C}{r}\left[1 - \frac{1}{(1+r)^T}\right]$ →

down.

$PV = \frac{C}{r}\left[1 - \frac{1}{(1+r)^T}\right]$



NPV. $= -50000 + \frac{5000}{1.1} + \frac{5000}{1.1^2} + \ldots\ldots\ldots$

NPV → is Posz. 1. 2. 3. > NPV is highest.

- **Qn**: Suppose David takes part in and wins a quiz show MU. MU offers David prize worth 1 million units, which will be paid in 20 equal installments of 50,000 units per year starting one year for now and with one new installement each year for the next 20 years. //
- Newspaper articles celebrate David saying he has become a millionaire overnight.
- However, his relative who studies Finance suggests that is not the case, and in fact the value of the prize is actually worth less than 500,000, given that the interest rate is 10% and is expected to remain at that rate foreover.
- Who is correct? $PV = \frac{50000}{0.1}\left(1 - \frac{1}{(1.1)^{20}}\right)$

$PV = \frac{50,000}{1.1} + \frac{50000}{1.1^2} + \dots + \frac{50000}{1.1^{20}}$

$= \frac{C}{r}\left[1 - \frac{1}{(1+r)^t}\right] = \frac{50000}{0.1}\left[1 - \frac{1}{(1.1)^{20}}\right]$

$= 500000\left[\frac{(1.1)^{20} - 1}{(1.1)^{20}}\right] \approx 425678 <$ t.

- **Qn**: Consider a coupon bond that will pay 1500 units each half-year for 5 years, plus an additional 100,000 units in 5 years.

How much should a bond be worth today, assuming that the interest rate is 5% per annum?

$1 + r_t = (1+r)^t$

Steps: find r for 6 months.

$1.05 = (1+r)^2 \therefore r = \sqrt{1.05} - 1 = 0.02469$

$= 2.47\%$

$\therefore r = 0.0247$

50% per year ∴ for 1500, $PV = \frac{1500}{0.0247}\left[1 - \frac{1}{(1+0.0247)^{10}}\right] = 13148.47$

6m rate for t 6m $1 + r_t = (1+r)^t$

$1 + 0.05 = (1+r)^2$

for 100000; $PV = \frac{100000}{1.055} = 78352.6$ ∴ $PV = 13148.5 + 78352.5$

a.m.

1) You bought 100 shares of LIC at the price of Rs 1,000 per share. LIC pays a quarterly dividend of Rs 5 per share. Suppose one year later the price of LIC will be Rs 1,100 per share. What is your rate of return of holding LIC shares?

$return = \frac{110000 + 2000 - 100000}{100000} - \frac{12000}{100000} = 0.12 = 12\%$

2) What should be the rate of interest per year so that your money doubles in 10 years?

$1 + 1 = (1+r)^{10} \quad \therefore (1+r)^{10} = 2$

$\therefore 10 \ln(1+r) = \ln 2 \quad \therefore \ln(1+r) = \frac{\ln 2}{10} = 0.0693.$

$\therefore 1 + r = e^{0.0693.} = 1.0717$

$\therefore r = 0.0717 = 7.17\%.$

3) A bank offers 6% annual interest on its savings account, but it credits interest quarterly into the savings accounts. What is the rate at which bank credits interest per quarter?

$1 + 0.06 = (1+r)^4 \quad \therefore 4 \ln(1+r) = \ln(1.06)$

$\therefore \ln(1+r) = \frac{0.0582}{4} = 0.01455$

$\therefore 1 + r = e^{0.01455} = 1.0146 \quad \therefore r = 0.0146 \times 100$

$1.46\% = 1.46\%$

4) A bond is priced at Rs 4,500 today. It has a maturity of 5 years. It pays Rs 500 at the end of each year beginning from next year and on maturity pays the face value of Rs 4,000. If prevailing interest rate is 10%, then Should you buy.

$NPV = -4500 + \frac{500}{1.1} + \frac{500}{1.1^2} + \dots + \frac{500}{1.1^5} + \frac{4000}{1.1^5}$

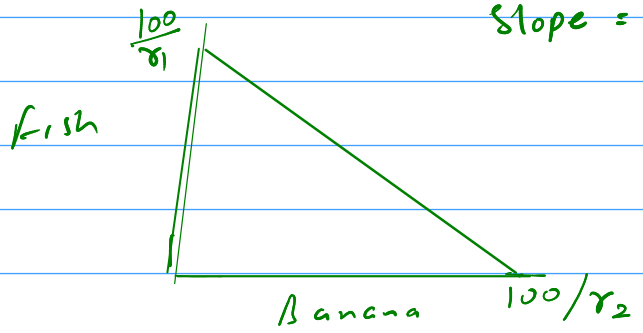
$= -4500 + \frac{500}{0.1}\left[1 - \frac{1}{(1.1)^5}\right] + 2483.68 = -2016.3 + 5000\left(\frac{1.1^5 - 1}{1.1^5}\right)$

$= -2016 + 1895.39 = -ive$, No.

100 Rs.

Fish $\rightarrow r_1$

Banana $\rightarrow r_2$.



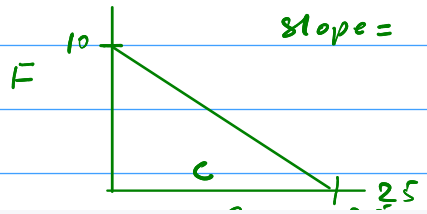
$$\text{Slope} = \frac{-y}{x} = -\frac{100/r_1}{100/r_2}.$$

5) If the interest rate were 6% per period, what would be the discount factor for cash received after 5 periods and cash received after 10 periods, respectively?

$\frac{1}{1.06^5}$, $\frac{1}{1.06^{10}}$.

6) A consumer has income of 100 units which he spends on a bundle of coconut and fish. The price of coconut is 4 units, and the price of fish is 10 units. What is the slope of the consumer's budget constraint when quantity of coconuts is taken on the horizontal axis and the quantity of fishes is taken on the vertical axis?

$100 \geq 4C + 10F$



Slope $= \frac{-10}{25} = -0.4$ ✓

7) You won a lottery and can receive either (1) Rs 60,000 today, or (2) Rs 10,000 one year from today plus Rs 25,000 two years from today plus Rs 35,000 three years from today. You plan to use the money to pay for your child's higher education in 15 years. Assume there is a constant interest rate throughout, then you should

- take the Rs 60,000 today because of the time value of money regardless of the interest rates.
- take option two because you get Rs 70,000 rather than Rs 60,000 regardless of the interest rates.
- take the Rs 60,000 today only if the interest rate is at least 16.67% per year.
- take the Rs 60,000 today if you can earn a constant 6.81% per year or more on your investments.

how to do this

take $r = 6.81\% = 0.0681$.

$$60000 = \frac{10000}{1+r} + \frac{25000}{(1+r)^2} + \frac{35000}{(1+r)^3}$$

then $\frac{10000}{1.0681} + \frac{25000}{1.0681^2} + \frac{35000}{1.0681^3}$ if this value < 60000, then take 60000

8) Suppose an asset sells at 250 units today and earns an annual interest rate of 12%. If interest is continuously compounded then what will be the value of asset after 3 years?

$A = P e^{rt}$. ✓

$250 \times e^{rt} = 358.33$

9) You invest in a project which assures a constant payment of Rs 15,000 per year. The payment starts from the current year and continues forever. If the prevailing interest rate is 15% then what is the amount of investment you need to make?

Perpetuity.

Read Carefully.

$\frac{C}{r}$

$PV = 15000 + \frac{15000}{0.15} = 115000$ ✓

10) You invest in a project which assures a payment of Rs 15,000 starting next year. The payment grows each year at a rate of 5% and lasts for 20 years. If the prevailing interest rate is 15% then what is the amount of investment you need to make?

$\frac{5000(1.05)}{1.15} + \frac{5000(1.05)^2}{1.15^2} + \dots$

$\left[\frac{C}{r-g}\left[1 - \frac{(1+g)^t}{(1+r)^t}\right]\right]$ ✓ remember this

$\therefore \frac{15000}{0.15-0.05}\left[1 - \left(\frac{1.05}{1.15}\right)^{20}\right] = 150000\left[0.605\right]$

≈ 90000

Annuity.

11) You purchased a coupon bond at Rs. 75,000 which pays Rs 5,000 annually for 10 years beginning from next year and a lump-sum Rs 100,000 in 10 years. Then,

- Your purchase is profitable if the prevailing interest rate is 8%.
- Your purchase is profitable if the prevailing interest rate is 9%.
- Your purchase is profitable if the prevailing interest rate is 10%.
- Your purchase is profitable if the prevailing interest rate is 11%.

$NPV = -75000 + \frac{5000}{r}\left[1 - \frac{1}{(1+r)^{10}}\right] + \frac{100000}{(1+r)^{10}}$

Put $r = 10, 9, 8$.

-ive -ive +ive ✓

12) How much money do I need to place into a bank account that pays a 2.7% annual interest rate in order to have Rs 50,000 at the end of 5 years? (Round to nearest integer)

$$\frac{50000}{(1+0.027)^5} = 43764$$

13) A zero-coupon bond pays no annual coupon interest payments. When it matures at the end of 7.5 years it pays out Rs. 10,000. If investors wish to earn 2.35% per year on this bond investment, what is the current price of the bond? (Round to nearest integer)

$$\frac{10000}{(1.0235)^{7.5}} = 8401.2$$

Imp.

14) You just invested Rs. 50,000 into an account that earns 7 percent compounded annually. At the end of each year, you can withdraw Rs. 5,000 (starting from next year). How many years can you continue to make withdrawals?

$$50000 = \frac{5000}{0.07}\left[1 - \frac{1}{1.07^t}\right] \Rightarrow \frac{1}{1.07^t} = 1 - \frac{7}{10} = \frac{3}{10}$$

Use same PV formula.

$$\therefore 1.07^t = \frac{10}{3}$$

$$\therefore t \log 1.07 = \log(3.33)$$

$$\therefore t = \frac{\log(3.33)}{\log(1.07)} = 17.78$$

15) A financial advisor tells you that you can make your child a crorepati if you just start saving early. You decide to put an equal amount each year into an investment account that earns 7.5% interest per year, starting the day your child is born. How much would you need to invest each year to accumulate one crore rupees for your child by the time he is 30 years old? (your last deposit will be made on his 29th birthday) **5 points**

RD, $FV = \frac{P}{r}\left[(1+r)^n - 1\right]$ Remember this.

$$\therefore 10000000 = \frac{P}{0.075}(1.075)^{30} - 1$$

$$\therefore P = \frac{10000000 \times 0.075}{(1.075)^{30} - 1} = 96712.2$$

16) You put Rs. 100,000 in an investment that returns 11% compounded monthly what amount would you have after 10 years? (rounded to nearest integer)

monthly rate. $1 + 0.11 = (1+r)^{12}$ $12 \ln(1+r) = \ln(1.11)$

$\therefore \ln(1+r) = 0.0087$

$1 + r = e^{0.0087} = 1.00873$

$\therefore r = 0.00873 = 0.873$

$$\therefore I = 100000 \times (1+0.00873)^{12} = 284055$$

Alternate way $r = \frac{11}{12} = 0.917$

$$\therefore I = 100000\,(1+0.00917)^{120} = 299003$$

17) Select the correct statements.

- [ ] The present value of an annuity increases as the discount rate increases. decreases
- [x] To evaluate or compare investment proposals, we must adjust the value of all cash flows to a common date. ✓
- [x] The future value of an annuity will increase if the interest rate goes up, but the present value of the same annuity will decrease as the interest rate goes up. ✓
- [ ] An investment earning simple interest is preferred over an investment earning compound interest. No.

Check.

18) Assume Alexa earns 100 units in period 1 and 120 units in period 2. She spends her earnings on consumption in period 1 and period 2. The prices of consumption in period 1 and period 2 are the same and equal to 1. If prevailing interest rate is 10% then,

- [ ] The maximum that she can consume in period 1 is 100 units. No, she can cons. adds $\frac{120}{1.1}$
- [x] The maximum that she can consume in period 2 is 230 units. Yes. $100(1.1) + 120 = 110 + 120 = 230$
- [x] If she wants to consume equal amount in both periods, she needs to borrow in period 1. Yes. Slope $= -(1+r) = -1.1$
- [ ] The slope of intertemporal budget constraint is -0.91.

19) Assume Alexa earns 100 units in period 1 and 120 units in period 2. She spends her earnings on consumption in period 1 and period 2. The price of consumption in period 1 is equal to 1. There is price inflation at rate = 5%. If prevailing interest rate is 10% then

Slope $= -\frac{(1+r)}{1+\pi} = \frac{-1.1}{1.05} = -1.0476$

- [x] The slope of intertemporal budget constraint is -1.048.
- [x] The maximum that she can consume in period 1 is 209.1 units. $\rightarrow 100 + \frac{120}{1.1} = 209.09$ (No effect of inflation)
- [ ] The maximum that she can consume in period 2 is 241.5 units. $100 \times 1.1 = 110 \rightarrow$ in next period, Price in next period = 1.05 $\therefore$ Consumption $= \frac{110}{1.05} + 120 = 224.76$
- [ ] She should save in the first period in order to consume the same amount in both periods.

20) Consider an asset that has a price of 100 units in period 1 and is expected to have a price of 115 units in period 2. There is no uncertainty and no information asymmetry. There are no dividends or other payments between periods 1 and 2. Select the correct statements.

- [ ] There exists no arbitrage opportunity if the prevailing interest rate is 12%.
- [x] There exists an arbitrage opportunity if the prevailing interest rate is 18%.
- [x] According to no arbitrage rule, the prevailing interest rate should be equal to 15%.
- [ ] According to no arbitrage rule, the prevailing interest rate should be less than 15%.

$115 = (1+r)100$

$\therefore r = \frac{115-100}{100} = \frac{15}{100} = 15\%$.

$\therefore$ At $r = 15\%$ no arbitrage.

if $r > 15\%$, arbitrage opportunity.

Week 3

Gordon Growth model. Share value : $\frac{\text{Dividend.}}{\text{rate - growth.}}$ $\frac{D}{r-g} \approx \left(\frac{C}{r-g}\right)$ perpetuity.

IRR is used when NPVs are similar.

$PI = \frac{PV(\text{future cash flows})}{-(\text{first cash flow})}$ [first cashflow has to be outflow]

If same NPV, higher IRR & higher PI projects may be preferred.

- **Example 1**: What is the two-year holding rate of return if the rate of interest is 30% in the first year and 20% in the second year?

$1+r = (1.3)(1.2) = 1.56 \quad \therefore r = 0.56 = 56\%$

- If one were earning a total three-year holding rate of return of 173% over the three year period, what would be the annualized rate of return?

$1+1.73 = (1+r)^3$

$\therefore 1+r = \sqrt[3]{2.73} = 1.397$

$\therefore r = 39.7\%$.

- **Example 3**: What annualized rate of return should one expect from a 100 unit investment today that promises a return of 240 units in 30 years?

$r = \frac{240-100}{100} = \frac{140}{100} = 1.4$

$\therefore 1+1.4 = (1+r)^{30}$.

$\therefore 1+r = (2.4)^{1/30} = 1.0296$.

$\therefore r = 0.0296 = 2.96\%$

$$\frac{5000}{1.1} + \frac{5000}{(1.1)^2} + \cdots\cdots$$

5000 5000 ..... 30 years.

$$\text{IRR.} = -50000 + \frac{5000}{1+\gamma} + \frac{5000}{(1+\gamma)^2} + \cdots\cdots \frac{5000}{(1+\gamma)^{30}} = 0$$

IRR $1+\gamma^3 \rightarrow (1+\gamma)^2$.

$$NPV = -50000 + \text{PV}$$

rate of interest is r.

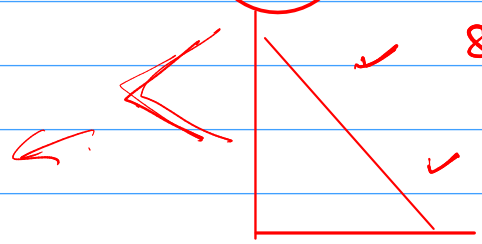


Price = 1)

1) Consume C_1 in first period, & C_2 in the 2nd.

2) Save C_1, $\longrightarrow$ $C_1(1+r) + C_2$

3) Borrow. $\frac{C_2}{1+r}$ + C_1 in first period.



Slope $= -(1+r)$ → formula.

Slope $= -\frac{(1+r)}{1+\pi}$ → inflation rate.

NPVs are similar NPV.

You can't find PI always.

$$PI. = \frac{PV.\ (\text{future cash flows})}{-(\text{first cash outflow})}$$

$$\frac{-5000 + \frac{5000}{1.1} + \frac{10000}{1.1^2} + \frac{5000}{1.13} + \frac{7000}{1.14}}{}$$

- **Example 4:** Suppose one has an investment that doubles in value in the first year, and then falls back to the original value.
- What would be the average rate of return?

1. $100 \longrightarrow 200 \quad \therefore r_1 = \frac{200-100}{100} = 1.$

2. $200 \longrightarrow 100 \quad \therefore r_2 = \frac{100-200}{200} = \frac{-100}{200} = -0.5.$

$\therefore$ Average return $= \frac{1-0.5}{2} = 0.25$

- **Example 5:** Consider an investment that gives a rate of return of $+20\%$ followed by a rate of return of -20%.
- This would have an average rate of return of 0%.
- However, in reality it would leave you will a loss.
- An investment of 1000 would become 1200 after first year, but drop down to 960 in the second year.
- The average rate of return of 0% does not accurately highlight this loss.
- If you calculate the annualized rate $r_2 = \sqrt{1-4\%} - 1 \approx -2.02\%$, it would show that one had a loss.

$100 \longrightarrow 120$

$120 \longrightarrow 96.$

Average return $= 0$, but actually loss.

$1 - 0.04 = (1+r)^2$

$\therefore r = \sqrt{0.96} - 1 \simeq -0.0202$

$= -2.02\%.$

- **Example 6:** Consider a project with an initial investment of 24 that pays 20 in one year and 16 in five years.
- Assume that the 1-year interest rate is 5% and the five-year annualized interest rate is 6% per annum.

$NPV = -24 + \frac{20}{1.05} + \frac{16}{1.06^5}$

CAs

1) Suppose SBI offers a home loan of Rs. 800,000 at 9% annual interest rate. A borrower takes the loan and agrees to repay in 20 years with equal monthly payments (beginning from next month). How much does the borrower need to pay each month? (use banking convention of multiplying monthly interest rate by 12 to get annual rate) (Rounded off to the nearest integer)

monthly rate $= \frac{9}{12} = 0.75\% = 0.0075$

$800000 = \frac{C}{0.0075}\left(1 - \frac{1}{1.0075^{240}}\right)$

$\therefore C = 6000 \times \frac{1.0075}{1.0075^{240}-1} = 7197.80. \approx 7198.$

2) Suppose SBI offers a home loan of Rs. 800,000 at 9% annual interest rate. A borrower takes the loan and agrees to repay in 20 years with equal monthly payments (beginning from next month). The borrower wins a lottery after 10 years (after 120 months) and wants to pay off the loan immediately. How much would he need to pay at that time? (use banking convention of multiplying monthly interest rate by 12 to get annual rate) (Rounded off to the nearest integer)

5 pol

First we find monthly installment $= 7198.$

Now after 10 years, years left $= 10.$

Now find pv of next 10 years.

$= \frac{7198}{0.0075}\left[1 - \frac{1}{(1.0075)^{120}}\right]$ ✓

3) ABC Limited has a forward-looking dividend yield of 3.5%. ABC Limited pays a dividend of Rs 50 per share this year. The dividend is expected to grow by 5% each year. What is the estimated appropriate cost of capital for ABC Limited?

forward looking dividend yield = $r - g$.

$\therefore r = 3.5 + 5 = 8.5\%$

4) Ram and Shyam are two friends, and both have income of 250 units per period. Ram is a person who tries to save as much as possible for the future while Shyam believes in consuming maximum in the current period. An investment opportunity appears which demands 250 units in the current period and returns 300 units in the next period. If the constant interest rate is 15% per period, then **5 poir**

$250 \times 1.15 = 287.50 < 300$

$\therefore$ irrespective of consumption behaviour, both should invest.

5) You invest in a project which pays Rs 500 next year and Rs 500 two years from today. The correct constant interest rate is 10% per year. But you make an error in your estimation and assume that project will pay Rs 600 in both periods, then

$NPV_1 = \frac{500}{1.1} + \frac{500}{1.1^2} = 867.16$

$NPV_2 = \frac{600}{1.1} + \frac{600}{1.1^2} = 1041.32$ $\quad \therefore NPV_2 = 1.2(NPV_1)$ = 20% more.

6) What is the IRR of a project that costs Rs 50,000 today and pays Rs 60,000 four years from today.?

$0 = -50000 + \frac{60000}{(1+r)^4} \Rightarrow (1+r)^4 = \frac{6}{5}$

$\therefore 4\ln(1+r) = \ln 1.2$

$\therefore \ln(1+r) = 0.04558$ $\quad \therefore r = e^{0.04558} - 1 = 0.0466 = 4.66\%$

IRR

7) What is the yield-to-maturity (YTM) of a two-year coupon bond that costs Rs 10,000 today and pays Rs 2,000 at the end of each year in coupon payments and also pays the value of Rs 8,000 at the end of the second year.

'Yield to maturity'

$0 = -10000 + \frac{2000}{1+r} + \frac{2000}{(1+r)^2} + \frac{8000}{(1+r)^2} = -10(1+r)^2 + 10 + 2(1+r) = 0$

$\Rightarrow -16 - 10r^2 - 20r + 16 + 2 + 2r = 0 \Rightarrow -10r^2 - 18r + 2 = 0$

$5r^2 + 9r - 1 = 0$

$r = \frac{-9 \pm \sqrt{81+20}}{10} = \frac{-9 + \sqrt{101}}{10} = 0.104 = 10.4\%$

8) Cobra Inc. is considering two different projects A and B for investment. Project A costs Rs 95,000 and is expected to generate Rs 65,000 in year one and Rs 75,000 is year two. Project B costs Rs 120,000 and is expected to generate Rs 64,000 in year one, Rs 67,000 in year two, Rs 56,000 in year three, and Rs 45,000 in year four. Cobra Inc.'s required rate of return for these projects is 10%. The net present value for Project A is (Rounded off to the nearest integer) **5 points**

$NPV_A = -95000 + \frac{65000}{1.1} + \frac{75000}{1.1^2} = 26074$

9) Cobra Inc. is considering two different projects A and B for investment. Project A costs Rs 95,000 and is expected to generate Rs 65,000 in year one and Rs 75,000 is year two. Project B costs Rs 120,000 and is expected to generate Rs 64,000 in year one, Rs 67,000 in year two, Rs 56,000 in year three, and Rs 45,000 in year four. Cobra Inc.'s required rate of return for these projects is 10%. The net present value for Project B is (Rounded off to the nearest integer) **5 points**

$NPV_B = -120000 + \frac{64000}{1.1} + \frac{67000}{1.1^2} + \frac{56000}{1.1^3} + \frac{45000}{1.1^4}$

$= 66362.8$

10) Cobra Inc. is considering two different projects A and B for investment. Project A costs Rs 95,000 and is expected to generate Rs 65,000 in year one and Rs 75,000 is year two. Project B costs Rs 120,000 and is expected to generate Rs 64,000 in year one, Rs 67,000 in year two, Rs 56,000 in year three, and Rs 45,000 in year four. Cobra Inc.'s required rate of return for these projects is 10%. The profitability index for Project A and Project B is respectively **5 points**

$$PI = \frac{PV(\text{future cash flows})}{-(\text{first cash flow})} = \frac{-90000 + A = 26074}{\therefore A = 116074}$$

$$\therefore PI_A = \frac{116074}{90000} = 1.289$$

Similarly $PI_B = (120000 + 66369)/120000 = 1.55$

11) Cobra Inc. is considering two different projects A and B for investment. Project A costs Rs 95,000 and is expected to generate Rs 65,000 in year one and Rs 75,000 is year two. Project B costs Rs 120,000 and is expected to generate Rs 64,000 in year one, Rs 67,000 in year two, Rs 56,000 in year three, and Rs 45,000 in year four. Cobra Inc.'s required rate of return for these projects is 10%. The internal rate of return for Project A is **5 points**

$$-95000 + \frac{65000}{1+r} + \frac{75000}{1+r^2} = 0 \Rightarrow -95(1+r)^2 + 65(1+r) + 75 = 0$$

Solve this

Similarly for Project B.

13) Vyom Corporation is considering a new system that will cost Rs 750,000. The system is expected to generate positive cashflows over t next four years in the amounts of Rs 350,000 in year one, Rs 325,000 in year two, Rs 150,000 in year three, and Rs 180,000 in year four. Ass the payment is equally distributed throughout a year. The company's required rate of return is 8%. What is the payback period of this systen

$$\text{payback} = \underset{1\text{ yr}}{350000} + \underset{1\text{ yr}}{325000} + \frac{75000}{150000} = 1 + 1 + 0.5 = 2.5$$

15) A project earns a total of 40% return in three years. If annual return in first years is 15%, and annual return in the second year is 12%. ther what is the annual rate of return in the third year?

$$1 + 0.4 = (1.15)(1.12)(1 + r_3)$$

$$\therefore 1 + r_3 = \frac{1.4}{(1.15)(1.12)} \quad \therefore r_3 = \frac{1.4}{(1.15)(1.12)} - 1$$

16) Suppose the share price of XYZ Corp. is Rs 150 today. The XYZ share price rises by 30% in the next period and then falls by 20% in the second period. What is the share price of XYZ Corp. after two periods?

first p $(150 + 0.3 \times 150) = 195$

s $195 - 0.2 \times 195 = 195 - 39 = 156$

17) Consider the following cashflows for projects and rates of return for different periods.

	Period 1	Period 2	Period 3	Period 4
Rate of return	**8%**	**12%**	**10%**	**8%**
Project 1	1000	1200	1300	1500
Project 2	800	1200	1400	1600
Project 3	0	2500	0	2500
Project 4	0	0	0	5000

Calculate NPVs for each.

$$NPV_1 = -3500 + \frac{1000}{1.08} + \frac{1200}{1.12} + \frac{1300}{1.3} + \frac{1500}{1.08^4}$$

If each project requires an initial investment of Rs 3500 in the current period, and you can choose only one project, then which of the project should be chosen according to the NPV capital budgeting rule?

20) Infinity Inc. has 100,000 shares outstanding which are trading at Rs 100 per share. Infinity Inc. is expected to earn Rs 500,000 as profit next year. The profit is expected to grow at 2% per annum thereafter. The firm faces a cost of capital of 8% per annum. If the firm pays a constant dividend of Rs 15 per year per share beginning from next year, then choose the correct statement(s) about Infinity Inc. **5 points**

- [x] According to the growing perpetuity formula on profit, the shares of Infinity Inc. are overpriced.
- [ ] According to the growing perpetuity formula on profit, the shares of Infinity Inc. are underpriced.
- [ ] According to the Gordon growth model, if the appropriate rate of return is 12%, then the shares of Infinity Inc. are overpriced.
- [x] According to the Gordon growth model, if the appropriate rate of return is 12%, then the shares of Infinity Inc. are underpriced.

Perpetuity $PV = \frac{C}{r-g} = \frac{500000}{0.08-0.02} = \frac{500000000}{6} = 83333333$ $\therefore$ per share price $= \frac{83333333}{100000} = 83.33$

Gordon Growth model $\frac{D}{P} = r - g$ $r = 12\%$ $g = 2\%$.

$\therefore P = \frac{15}{0.10} = 150$ $\therefore$ price is underpriced.

Week 4

- **Duration**: payment-weighted average of all "payout times", and is equal to

$$\frac{\sum_t C_t t}{\sum_t C_t}$$

Maturity - 1 is last time

$\frac{50000}{1.1^5} + \frac{100000}{1.1^{10}}$

- Another commonly used measure is **Macauley Duration**.
- Instead of weighted average of cashflows, Macauley Duration is a weighted average of present values:

$$\frac{\sum_t PV(C_t)\, t}{\sum_t PV(C_t)}.$$

dura

$$CPI = \frac{\sum_i p_C^i q_B^i}{\sum_i p_B^i q_B^i} \times 100$$

- Nominal GDP is the value of GDP using current prices.

real

- Real GDP measures the value of production using constant prices (price in some base year).

- GDP deflator (P_t) for a particular year, is defined as the ratio of nominal GDP to real GDP for that year:

$$P_t = \frac{\text{Nominal GDP in year t}}{\text{Real GDP in year t}}$$

- Interest rates expressed in units of currency are called **nominal interest rates**.

- Interest rates expressed in terms of baskets of goods are called **real interest rates**.

$$(1+\rho) = \frac{1+r}{1+\pi}$$

$$1 + r_t = \frac{(1 + i_t)}{(1 + \pi^e_{t+1})}.$$

PYQs

$\frac{\sum C_i t_i}{\sum C_i} = \frac{500 \times 5 + 1000 \times 10}{500 + 1000}$

Consider a bond which pays 500 units in 5 years and 1000 units in 10 years. What is the maturity and duration of the bond, respectively?

OPTIONS:

- ○ 10 years, 10 years
- ○ 5 years, 5 years
- ✓ 10 years, 8.33 years
- ○ 5 years, 8.33 years

maturity → last day of payment. ∴ 10 years.

$\text{duration} = \frac{\sum C_t t}{\sum C_t} = \frac{500 \times 5 + 1000 \times 10}{1500} = \frac{12500}{1500} = \frac{25}{3} = 8.33\%$

$\frac{(2018) - (2017)}{(2017)}$

The table below relates to the economy of Galaland, where the typical consumer's market basket consists of 2 clothes and 3 burgers.

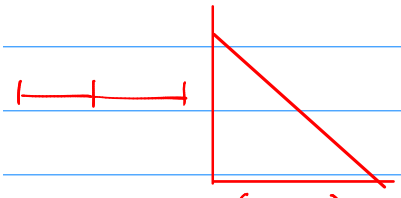
Year	Price of a cloth	Price of a burger
2017	Rs 400	Rs 30
2018	Rs 300	Rs 50
2019	Rs 325	Rs 70

$CPI(2018) = \frac{(2 \times 300 + 3 \times 50) - (2 \times 400 + 3 \times 30)}{2 \times 400 + 3 \times 30}$

$= \frac{750 - 890}{890} = -\frac{140}{890} = -0.157 = -15.7\%$

If the base year is 2017, then the economy's inflation rate (as measured by CPI) in 2018 is −15.7%.

Assume Shyam earns Rs. 10000 in period 1 and Rs 12000 in period 2. He spends his earnings on consumption in period 1 and period 2. The prices of consumption in period 1 and period 2 are the same (equal to 1). If prevailing interest rate is 10% then the slope of the intertemporal budget constraint is



$= -(1+r) = -1.1$

inter slop $-(1+r)$

The current average price level of goods in the economy of Godzilla is 80 units. The prevailing nominal interest rate is 10%. If expected price level in the next period is 90 units, then the real interest rate in the economy is (approximately)

$1 + \pi = \frac{1+r}{1+\pi}$ $\quad r = 0.1$

$\pi = \frac{90-80}{80} = \frac{1}{8} = 0.125$

$\therefore 1 + \pi = \frac{1.1}{1.125} = 0.977$ $\quad \therefore \pi = 0.977 - 1 = -0.0233$

$= -2.33\% \simeq -2.5\%$

-2.5%

A company with a AAA bond rating will command a higher interest rate on its bonds than a company with lesser BBB bond rating. False.

False

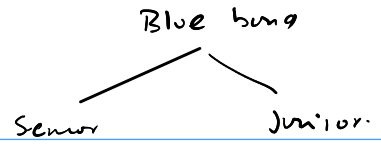
AAA – less risk – low interest rate.

BBB – higher risk – higher " "

↑ AAA → lower risk → lower interest

↓ BBB → higher risk → higher interest rate.

Your portfolio consists of two blue bonds: each has a face value of 100 units and a probability of default $p_b = 20\%$. The blue bonds are perfectly correlated with each other ($\gamma = 1$). Through securitization, you are able to create two new bonds with face values: a senior bond and a junior bond. When none of the blue bond default both new bonds are paid face value of 100. If any one of the blue bond defaults, the senior bond is paid the face value of 100 while the junior bond suffers the loss and is paid 0. In case of both the blue bonds defaulting, none of the new bonds pay off. What is the probability that the senior bond results in default?



$\gamma = 1$ ∴ Correlated. ∴ P (default) remains same : 0.2 = 20%.

of both the blue bonds defaulting, none of the new bonds pays off. What is the expected payoff of the junior bond?

Perfectly Correlated ⟨ either both default / or none defaults.

O × O = 20%

✓ P (none defaults) = $1 - P_d$ = 1 − 0.2 = 0.8 ↳ Junior gets 100

P (both default) = 0.2 ✓ Junior gets 0.

∴ ✓ Expected payoff (junior) : 0.8 × 100 + 0.2 × 0 = 80. ✓

A basis point is equal to

full point = 1%.

bons point : 0.01 %.

What is the term for a graphical representation of the relationship between interest rates and the maturities of debt securities? (week 4)

Options :

6406533355837. GDP growth

6406533355838. Maturity chart

6406533355839. Yield curve ✓ (week 4)

6406533355840. Inflationary expectations

What should be the rate of interest per year so that your money triples in 15 years?

Options :

$1 + r_t = (1+r)^t$

$1 + 2 = (1+r)^{15}$ $\ln(1+r) = \frac{\ln 3}{15} = 0.07324$

$1 + 2 = (1+r)^t$ log ×

$\therefore 1 + r = e^{0.07324} = 1.07598.$

$r = e^{\frac{\ln 3}{15}} - 1$

$\therefore r = 0.07598 \approx 7.6\%$ ✓

An investment is expected to yield Rs 3000 in three years, Rs 5000 in 5 years, and Rs 7000 in seven years. What is the present value of this investment if the constant rate of interest is 10% annually? (Rounded off to the nearest integer)

Options :

6406533355845. Rs 15000

6406533355846. Rs 11598

6406533355847. Rs 10000

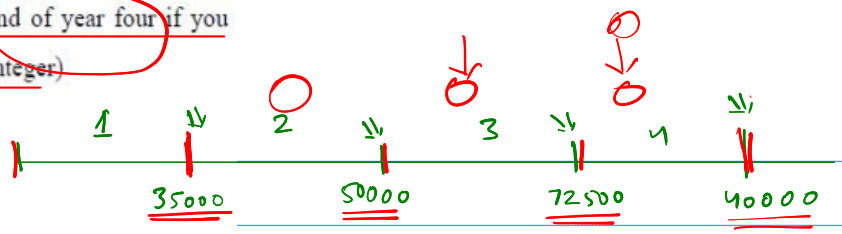
6406533355848. Rs 8951 ✓

$PV = \frac{3000}{1.1^3} + \frac{5000}{1.1^5} + \frac{7000}{1.1^7} = 2253.94 + 3104.6 + 3592.1$ ✓

$= 8950.64 \approx 8951$ ✓

You have been depositing money at the end of each year into an account drawing 8% interest annually. What is the balance in the account at the end of year four if you deposited the following amounts? (Rounded off to the nearest integer)

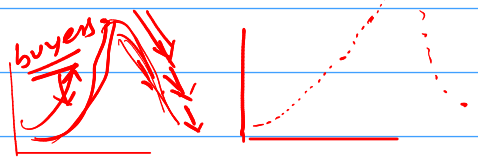
Year	End of year deposit
1	Rs 35,000
2	Rs 50,000
3	Rs 72,500
4	Rs 40,000



$\therefore 35000(1.08)^3 + 50000(1.08)^2 + 72500(1.08) + 40000$

$47617.6 + 58320 + 78300 + 40000$

224237 ✓



Which of the following statements about bubbles is correct?
Options :
6406533355853. A bubble occurs when the fundamental value of a share rises too quickly. False
6406533355854. A bubble is less likely to occur in a market where people can easily switch from buying to selling. False
6406533355855. Momentum trading strategies make bubbles more likely to occur. True.
6406533355856. A bubble can never occur in financial markets as trades take place very frequently False

Suppose an asset earns an annual interest rate of 8% which is continuously compounded. If asset sells for 100 units today, what will be its value after 2 years?
Options :
6406533355857. 108.34
6406533355858. 116.64
6406533355859. 117.35
6406533355860. 121.65

$100 e^{rt} = 100 \times e^{0.08 \times 2} = 100 \times e^{0.16}$

$= 117.35$

What is the present value of a cash flow stream that pays Rs 3000 next year, grows at a rate of 3% per year thereafter and faces a cost of capital of 9% per year? (? no. of years)
Options :
6406533355861. Rs 100,000 ✓
6406533355862. Rs 50,000
6406533355863. Rs 33,333
6406533355864. Rs 9,000

$= \frac{C}{r-g}$ (Perpetuity) $= \frac{3000}{0.09-0.03} = \frac{3000}{0.06} = 50000$ ✓

Question Label : Multiple Choice Question
You discover an antique in your attic that your father purchased 30 years ago for Rs 400. You auction it and receive Rs 80,000 for this item today. What is the annualized rate of return that you earned?
Options :
6406533355865. 6.67%
6406533355866. 11.67%
6406533355867. 19.32% ✓
6406533355868. 26.54%

t, C_0

rate of interest $= \frac{80000-400}{400} = \frac{79600}{400} = 199 = 199.$

$\therefore 1+199 = (1+r)^{30} \quad \therefore r = e^{\left(\frac{\ln 200}{30}\right)} - 1$ ✓

$r = 0.1932 = 19.32\%$

You believe in the power of compounding and decide to save Rs 100 per day by avoiding the junk foods. You deposit the Rs 100 at the end of each day in a bank account that pays 8% annual interest compounded daily. How much money will you have in 5 years, assuming 365 days per years?
(use banking convention: daily interest rate = r/365) (Rounded off to the nearest integer)

$r_d = \frac{0.08}{365}$

Options :
6406533355869. Rs 182,500
6406533355870. Rs 214,131
6406533355871. Rs 224,365 ✓
6406533355872. Rs 254,689

daily interest rate : $\frac{0.08}{365 \times 10^4} = \frac{2.1917}{10^9} = 0.00021917$

no. of days = $5 \times 365 = 1825$ days.

R.D.

$$A = \frac{100}{r}\left[(1+r)^t - 1\right] = \frac{100}{0.00021917} \times \left[(1.00021917)^{1825} - 1\right]$$

$$456266.66 \times [0.4917]$$

$$224363.45$$

Perpetuity $P = \frac{D}{r-g}$

The LIC stock is trading at Rs 400 today. LIC is expected to pay dividend of Rs 30 per share next year. If the growth rate of LIC dividend is 4% every year and appropriate rate of return is 10% per year, then according to the Gordon growth model
Options :
6406533355873. The LIC stock is underpriced, and an investor should buy LIC shares.
6406533355874. The LIC stock is overpriced, and an investor should buy LIC shares.
6406533355875. The LIC stock is underpriced, and an investor should sell LIC shares.
6406533355876. The LIC stock is overpriced, and an investor should sell LIC shares.

$$P = \frac{D}{r-g} = \frac{30}{0.1 - 0.04} = \frac{3000}{0.06} = 500.$$

∴ Share is underpriced.

Suppose an investment cost 5,000 today and pays off Rs 3,000 in the next year and Rs 4,000 in two years. What is the internal rate of return on this investment?
Options :
6406533355877. 24.34% ✓
6406533355878. 40%
6406533355879. 33.33%
6406533355880. 18.56%

$$0 = -5000 + \frac{3000}{1+r} + \frac{4000}{(1+r)^2}$$

$$-5(1+r)^2 + 3(1+r) + 4 = 0.$$

$$5(1+r)^2 - 3(1+r) - 4 = 0.$$

$$5r^2 + 5 + 10r - 3 - 3r - 4 = 0$$

$$5r^2 + 7r - 2 = 0$$

$r^3 \to 3.$

real roots 2

$$\therefore r = \frac{-7 \pm \sqrt{49 + 40}}{10} = \frac{\sqrt{89} - 7}{10} = 0.2433 = 24.33\%$$

Which of the following statements is correct regarding internal rate of return (IRR)?
Options :
6406533355881. The internal rate of return can be computed for every type of project. No.
6406533355882. The internal rate of return can never be zero. No.
6406533355883. Every project has a unique internal rate of return. No.
6406533355884. The IRR capital budgeting rule may give different answer than the NPV capital budgeting rule. Yes. ✓

There is a project which requires an initial investment of 500 units today. This project pays off 500 units after 5 years, 500 units after 10 years and 500 units after 15 years. If the prevailing constant interest rate in the economy is 10% then what is the profitability index of the project?

Options :

6406533355885. 122.93

6406533355886. 3.56

6406533355887. 1.25

6406533355888. 0.80

PI = PV (future cash inflows) / (first cash outflow)

Cash outflow.

PV (future Cash flows) $= \frac{500}{1.15} + \frac{500}{1.1^{10}} + \frac{500}{1.1^{15}}$

$\therefore 500\,(0.62 + 0.3855 + 0.2393) = 500 \times 1.2448$

$\therefore PI = \frac{500 \times 1.25}{-(-500)} = 1.25$ $= 500 \times 1.25$

Consider the following three projects which have initial cost of Rs 1000 today:

Project 1: It pays out Rs. 500 next year and Rs 1000 two years from today.

Project 2: It pays Rs. 100 starting next year and pays the same amount for 50 years.

Project 3: It pays Rs. 4000 five years from today. There are no interim payments.

Assume that the prevailing interest rate is 5%. Which project has the least payback period?

Pay 1yr. 10yr. Pay 10. After 5 yr. 2..

Options :

6406533355889. Project 1

6406533355890. Project 2

6406533355891. Project 3

6406533355892. All three projects have same payback period.

Payback 1 $= 500 \quad \frac{500}{1000} = 1.5$ ∴ lowest Project 1

Payback 2 $= 100 + 100 + 100 + 100 + \ldots = 10$

Payback 3 $= 5$

A project earns a total of 50% return in two years. If annual return in the first year is 30%, then what is the annual return in the second year? (Round off to nearest integer)

Options :

6406533355893. 30%

6406533355894. 25%

6406533355895. 20%

6406533355896. 15% ≈

$(1+r_t) = (1+r_1)(1+r_2)(1+r_3)\ldots$

$1 + 0.5 = (1+0.3)(1+r_2)$

$\therefore 1 + r_2 = \frac{1.5}{1.3} = 1.1538$

$\therefore r_2 = 0.1538 \simeq 15\%$.

What is the annualized rate of return of an investment of 100 units that promises to return 200 units in 10 years?

Options :

6406533355897. 10%

6406533355898. 7.18%

6406533355899. 5.97%

6406533355900. 3.76%

$1 + 1 = (1 + r)^{10}$.

$\therefore r = e^{\left(\frac{\ln 2}{10}\right)} - 1 = 0.0717 \simeq 7.18\%$

tip → always ln.

A financial advisor tells you that you can become a millionaire if you start investing early. You start to deposit 25,000 units every year in a savings account which earns 8% annual interest. What is the earliest year at which your account balance reaches a million units? (Assume you don't make any withdrawals in between)

Options :

6406533355901. 40

6406533355902. 25

6406533355903. 19

6406533355904. 15

RD.

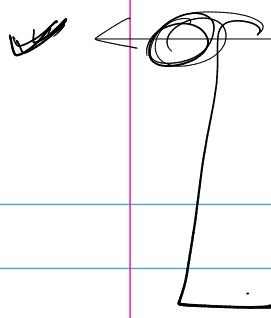
$A = \frac{2500}{r}\left[(1+r)^t - 1\right]$

$1 + \frac{100000 \times 0.08}{2500 \times 100} = (1.08)^t \Rightarrow 1.08^t = 1 + \frac{16}{5} = 1 + 3.2 = 4.2$

$\therefore t \ln 1.08 = \ln 4.2$

$t = \frac{\ln 4.2}{\ln 1.08} = \frac{1.435}{0.0769} = 18.64 \approx 19.$

Recurrent. $= \frac{C}{r}\left[(1+r)^t - 1\right]$.



→ NPV.

→ IRR. (rate of interest)

→ P.I. → NPV and I.

→ payback (how fast I get my capital